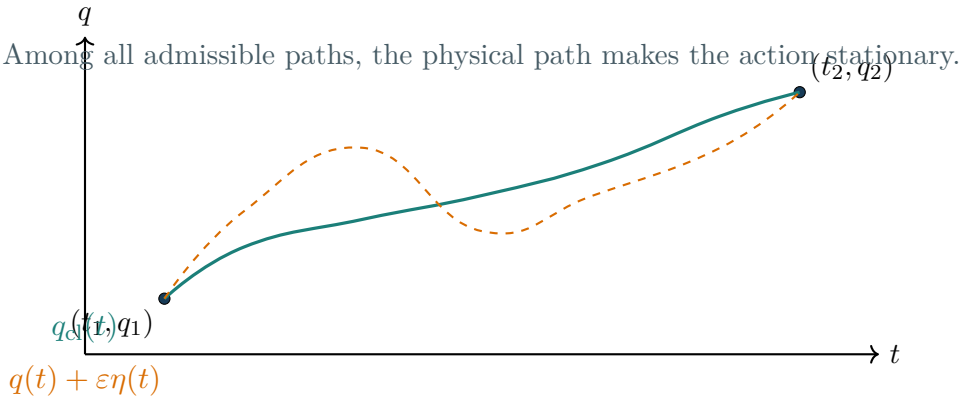
The Principle of Least Action

Advanced Lecture Notes for Theoretical Physicists

Prepared as a self-contained lecture manuscript



Guiding idea

The equations of motion can be encoded in one scalar object, the *action*. For a mechanical system with generalized coordinates $q^i(t)$,

$$\mathcal{S}[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\,\mathrm{d}t.$$

The physical trajectory is not usually the path of smallest length, smallest time, or smallest energy. It is a path for which the first variation of the action vanishes:

$$\delta\mathcal{S} = 0.$$

This is why modern authors often prefer the term *principle of stationary action* rather than *principle of least action*.

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1 What the action principle is really saying

The principle of least action is one of the most compact formulations of theoretical physics. It appears in classical mechanics, optics, field theory, general relativity, quantum mechanics, statistical field theory, and modern gauge theory. Its power is not that it gives a clever alternative way to solve Newton's equations. Its power is that it reveals the geometric and symmetry structure behind the laws of motion.

In its simplest mechanical form, the principle says:

$$\delta \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t = 0.$$

The object L is the Lagrangian. For many elementary mechanical systems,

$$L = T - V,$$

where T is kinetic energy and V is potential energy. But this formula is not the definition of a Lagrangian in general. In relativistic mechanics, field theory, and gauge theory the Lagrangian is better understood as the object whose variational derivative gives the equations of motion and whose symmetries encode conservation laws.

1.1 Stationary does not mean minimal

A common misconception is that the physical path always minimizes the action. In many simple short-time problems this is true. However, in general the action can have a local minimum, a local maximum, or a saddle point.

The condition is not

$$\mathcal{S}[q_{\mathrm{cl}}] \leq \mathcal{S}[q]$$

for all nearby paths. The universal condition is

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} \mathcal{S}[q_{\mathrm{cl}} + \varepsilon\eta]\Big|_{\varepsilon=0} = 0$$

for every allowed variation η satisfying the required boundary conditions.

Terminology

The historical phrase "least action" is useful, but technically the fundamental equation is *stationary action*. The first variation vanishes. The second variation decides whether the stationary point is a minimum, maximum, or saddle.

1.2 Why a variational principle is powerful

There are at least five reasons why the action formulation is central in theoretical physics.

1. It handles generalized coordinates naturally. Coordinates can be angles, distances, fields, maps, gauge potentials, or metrics.
2. It incorporates constraints cleanly, especially holonomic constraints and gauge constraints.
3. It exposes symmetries. Noether's theorem connects continuous symmetries with conserved quantities.
4. It generalizes from particles to fields by replacing paths $q(t)$ with field configurations $\phi(x)$.
5. It is the classical limit of the quantum path integral, where amplitudes involve $e^{i\mathcal{S}/\hbar}$.

2 Calculus of variations in one dimension

2.1 Functionals and variations

A function takes a number and returns a number. A functional takes a function and returns a number. The action is a functional because it assigns to each path $q(t)$ a number:

$$q(t) \mapsto \mathcal{S}[q].$$

Let

$$\mathcal{S}[q] = \int_{t_1}^{t_2} L(q(t), \dot{q}(t), t)\,\mathrm{d}t.$$

We perturb the path by a small parameter ε:

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

where $\eta(t)$ is the variation. If the endpoints are fixed, then

$$\eta(t_1) = \eta(t_2) = 0.$$

First variation

The first variation of $\mathcal{S}$ in the direction η is

$$\delta\mathcal{S}[q;\eta] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\mathcal{S}[q+\varepsilon\eta]\bigg|_{\varepsilon=0}.$$

A path q_{cl} is stationary if

$$\delta\mathcal{S}[q_{\mathrm{cl}};\eta] = 0$$

for every allowed variation η.

2.2 Derivation of the Euler-Lagrange equation

Let

$$\mathcal{S}[q_\varepsilon] = \int_{t_1}^{t_2} L(q+\varepsilon\eta, \dot{q}+\varepsilon\dot{\eta}, t)\,\mathrm{d}t.$$

Differentiate with respect to ε and evaluate at $\varepsilon = 0$:

$$\delta\mathcal{S} = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta}\right)\,\mathrm{d}t.$$

The second term contains $\dot{\eta}$. We integrate it by parts:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}}\dot{\eta}\,\mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}}\eta\right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)\eta\,\mathrm{d}t.$$

For fixed endpoints, $\eta(t_1) = \eta(t_2) = 0$, so the boundary term vanishes. Hence

$$\delta\mathcal{S} = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)\right]\eta(t)\,\mathrm{d}t.$$

If this is zero for every admissible η, then the bracket must vanish:

$$\boxed{\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) = 0.}$$

This is the Euler-Lagrange equation.

Euler-Lagrange equation

For the action

$$\mathcal{S}[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\,\mathrm{d}t,$$

with fixed endpoint variations, every stationary path satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0.$$

2.3 The fundamental lemma

The key step in the derivation is the following fact. If a continuous function $F(t)$ satisfies

$$\int_{t_1}^{t_2} F(t)\eta(t)\,\mathrm{d}t = 0$$

for every smooth function η that vanishes at the endpoints, then

$$F(t) = 0$$

for all t in the interval. This is called the fundamental lemma of the calculus of variations.

Intuitively, if F were positive somewhere, we could choose η positive and localized near that place, making the integral positive. Therefore the only way the integral can vanish for all local tests η is for F itself to vanish.

2.4 Several generalized coordinates

For n generalized coordinates $q^1, \ldots, q^n$, the action is

$$\mathcal{S}[q] = \int_{t_1}^{t_2} L(q^1, \ldots, q^n, \dot{q}^1, \ldots, \dot{q}^n, t)\,\mathrm{d}t.$$

Independent variations give one equation for each coordinate:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i}\right) - \frac{\partial L}{\partial q^i} = 0, \qquad i = 1, \ldots, n.}$$

In compact notation:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i}\right) = \frac{\partial L}{\partial q^i}.$$

The index i labels a coordinate, not a power.

3 Elementary examples with complete derivations

3.1 Free particle

Consider a particle of mass m moving on a line. The Lagrangian is

$$L = \frac{1}{2}m\dot{x}^2.$$

Then

$$\frac{\partial L}{\partial x} = 0, \qquad \frac{\partial L}{\partial \dot{x}} = m\dot{x}.$$

The Euler-Lagrange equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(m\dot{x}) = 0.$$

Therefore

$$m\ddot{x} = 0,$$

so

$$x(t) = At + B.$$

If the endpoint data are

$$x(t_1) = x_1, \qquad x(t_2) = x_2,$$

then

$$A = \frac{x_2 - x_1}{t_2 - t_1}, \qquad B = x_1 - \frac{x_2 - x_1}{t_2 - t_1} t_1.$$

Thus

$$\boxed{x_{\mathrm{cl}}(t) = x_1 + \frac{x_2 - x_1}{t_2 - t_1}(t - t_1).}$$

The stationary path is a straight line in spacetime.

Classical action of a free particle

Along the classical path,

$$\dot{x}_{\mathrm{cl}} = \frac{x_2 - x_1}{t_2 - t_1}.$$

Therefore

$$\mathcal{S}_{\mathrm{cl}} = \int_{t_1}^{t_2} \frac{1}{2} m \left(\frac{x_2 - x_1}{t_2 - t_1} \right)^2 \mathrm{d}t = \frac{m(x_2 - x_1)^2}{2(t_2 - t_1)}.$$

This expression is important in the semiclassical propagator of quantum mechanics.

3.2 Particle in a potential

Let

$$L = T - V = \frac{1}{2} m\dot{x}^2 - V(x).$$

Then

$$\frac{\partial L}{\partial x} = -\frac{\mathrm{d}V}{\mathrm{d}x}, \qquad \frac{\partial L}{\partial \dot{x}} = m\dot{x}.$$

The Euler-Lagrange equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}(m\dot{x}) + \frac{\mathrm{d}V}{\mathrm{d}x} = 0.$$

Thus

$$m\ddot{x} = -\frac{\mathrm{d}V}{\mathrm{d}x}.$$

This is Newton's second law with force

$$F(x) = -\frac{\mathrm{d}V}{\mathrm{d}x}.$$

3.3 Harmonic oscillator

For a harmonic oscillator,

$$V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2 x^2,$$

where $\omega^2 = k/m$. The Lagrangian is

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2 x^2.$$

The Euler-Lagrange equation is

$$m\ddot{x} + m\omega^2 x = 0,$$

or

$$\boxed{\ddot{x} + \omega^2 x = 0.}$$

The general solution is

$$x(t) = A\cos\omega t + B\sin\omega t.$$

For endpoint data $x(0) = x_i$ and $x(T) = x_f$, assuming $\sin\omega T \neq 0$, we find

$$A = x_i, \qquad B = \frac{x_f - x_i\cos\omega T}{\sin\omega T}.$$

Hence

$$\boxed{x_{\mathrm{cl}}(t) = x_i\cos\omega t + \frac{x_f - x_i\cos\omega T}{\sin\omega T}\sin\omega t.}$$

Classical action of the harmonic oscillator

For the harmonic oscillator, along any solution of the equation of motion,

$$\ddot{x} = -\omega^2 x.$$

Compute

$$\frac{\mathrm{d}}{\mathrm{d}t}(x\dot{x}) = \dot{x}^2 + x\ddot{x} = \dot{x}^2 - \omega^2 x^2.$$

Thus the Lagrangian on shell is

$$L_{\mathrm{on-shell}} = \frac{1}{2}m(\dot{x}^2 - \omega^2 x^2) = \frac{1}{2}m\frac{\mathrm{d}}{\mathrm{d}t}(x\dot{x}).$$

Therefore

$$\mathcal{S}_{\mathrm{cl}} = \frac{1}{2}m[x\dot{x}]_0^T.$$

Using the endpoint solution, one obtains

$$\boxed{\mathcal{S}_{\mathrm{cl}} = \frac{m\omega}{2\sin\omega T}\left[(x_f^2 + x_i^2)\cos\omega T - 2x_i x_f\right].}$$

This formula becomes singular at $\sin\omega T = 0$ because the boundary value problem is then degenerate.

3.4 Stationary action can fail to be a minimum

For the harmonic oscillator, the second variation is

$$\delta^2\mathcal{S} = \frac{m}{2}\int_0^T (\dot{\eta}^2 - \omega^2\eta^2)\,\mathrm{d}t.$$

Choose

$$\eta(t) = \sin\left(\frac{\pi t}{T}\right).$$

Then

$$\int_0^T \dot{\eta}^2 \,\mathrm{d}t = \frac{\pi^2}{2T}, \qquad \int_0^T \eta^2 \,\mathrm{d}t = \frac{T}{2}.$$

Hence

$$\delta^2 \mathcal{S} = \frac{m}{4}\left(\frac{\pi^2}{T} - \omega^2 T\right).$$

This is positive if $T < \pi/\omega$ and negative if $T > \pi/\omega$. Therefore the stationary trajectory is a minimum only for sufficiently short time intervals. At longer times it becomes a saddle.

4 Boundary terms and natural boundary conditions

In deriving the Euler-Lagrange equation, we usually impose fixed endpoints. If endpoints are not fixed, the boundary term does not vanish automatically.

For one coordinate,

$$\delta\mathcal{S} = [p\eta]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}}\right)\eta \,\mathrm{d}t,$$

where

$$p = \frac{\partial L}{\partial \dot{q}}$$

is the canonical momentum.

If the final endpoint is free, then $\eta(t_2)$ can be arbitrary. Stationarity requires

$$p(t_2) = 0$$

unless an additional boundary term is supplied.

Boundary data are part of the variational problem

One does not only specify an action. One must also specify the class of allowed paths and the boundary conditions. Changing boundary conditions can change the physical equations at the boundary, even if the bulk Euler-Lagrange equations are unchanged.

4.1 Adding total time derivatives

Suppose two Lagrangians differ by a total derivative:

$$L'(q, \dot{q}, t) = L(q, \dot{q}, t) + \frac{\mathrm{d}}{\mathrm{d}t}F(q, t).$$

Then

$$\mathcal{S}'[q] = \mathcal{S}[q] + F(q(t_2), t_2) - F(q(t_1), t_1).$$

For fixed endpoints, the added term is constant under variations, so

$$\delta\mathcal{S}' = \delta\mathcal{S}.$$

Thus L and L' produce the same Euler-Lagrange equations.

This simple fact is crucial in gauge theory and in Noether's theorem, where symmetries may change the Lagrangian by a total derivative rather than leaving it strictly invariant.

5 Generalized coordinates and geometry of configuration space

A mechanical system with n degrees of freedom has a configuration space Q. A point $q \in Q$ represents an instantaneous configuration. A motion is a curve

$$q : I \to Q.$$

In local coordinates q^i, a tangent vector is represented by $\dot{q}^i$. The Lagrangian is a function on the tangent bundle:

$$L : TQ \times \mathbb{R} \to \mathbb{R}.$$

5.1 Kinetic energy as a metric

Many mechanical systems have kinetic energy of the form

$$T = \frac{1}{2} g_{ij}(q) \dot{q}^i \dot{q}^j.$$

Here $g_{ij}(q)$ is a positive definite metric on configuration space. The Lagrangian

$$L = \frac{1}{2} g_{ij}(q) \dot{q}^i \dot{q}^j - V(q)$$

leads to equations that include Christoffel-symbol-like terms.

Compute

$$\frac{\partial L}{\partial \dot{q}^k} = g_{kj} \dot{q}^j,$$

and

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{q}^k} = g_{kj} \ddot{q}^j + \frac{\partial g_{kj}}{\partial q^\ell} \dot{q}^\ell \dot{q}^j.$$

Also,

$$\frac{\partial L}{\partial q^k} = \frac{1}{2} \frac{\partial g_{ij}}{\partial q^k} \dot{q}^i \dot{q}^j - \frac{\partial V}{\partial q^k}.$$

Therefore

$$g_{kj} \ddot{q}^j + \frac{\partial g_{kj}}{\partial q^\ell} \dot{q}^\ell \dot{q}^j - \frac{1}{2} \frac{\partial g_{ij}}{\partial q^k} \dot{q}^i \dot{q}^j + \frac{\partial V}{\partial q^k} = 0.$$

After raising an index with g^{mk}, this can be written as

$$\ddot{q}^m + \Gamma^m_{ij} \dot{q}^i \dot{q}^j = -g^{mk} \frac{\partial V}{\partial q^k},$$

where

$$\Gamma^m_{ij} = \frac{1}{2} g^{mk} \left(\frac{\partial g_{kj}}{\partial q^i} + \frac{\partial g_{ki}}{\partial q^j} - \frac{\partial g_{ij}}{\partial q^k} \right).$$

For $V = 0$, the motion is geodesic motion on configuration space.

6 Constraints and Lagrange multipliers

6.1 Holonomic constraints

A holonomic constraint has the form

$$f_a(q, t) = 0, \qquad a = 1, \ldots, r.$$

There are two common ways to handle such constraints.

1. Solve the constraints explicitly and use independent generalized coordinates.
2. Introduce Lagrange multipliers $\lambda^a(t)$ and extremize the extended action.

The extended Lagrangian is

$$L_{\text{ext}}(q, \dot{q}, \lambda, t) = L(q, \dot{q}, t) + \lambda^a f_a(q, t).$$

Variation with respect to λ^a gives the constraints:

$$f_a(q, t) = 0.$$

Variation with respect to q^i gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = \lambda^a \frac{\partial f_a}{\partial q^i}.$$

Depending on sign convention, the multiplier term may appear with the opposite sign. The multipliers encode constraint forces.

6.2 Example: particle constrained to a sphere

Consider a particle in $\mathbb{R}^3$ constrained to a sphere of radius R:

$$f(\boldsymbol{x}) = \boldsymbol{x}^2 - R^2 = 0.$$

The free-particle Lagrangian is

$$L = \frac{1}{2}m\dot{\boldsymbol{x}}^2.$$

Use

$$L_{\text{ext}} = \frac{1}{2}m\dot{\boldsymbol{x}}^2 + \lambda(\boldsymbol{x}^2 - R^2).$$

The Euler-Lagrange equation for x^i is

$$m\ddot{x}^i = 2\lambda x^i,$$

or

$$m\ddot{\boldsymbol{x}} = 2\lambda \boldsymbol{x}.$$

The acceleration is radial, as expected for motion on a sphere with no external tangential force.

To solve for λ, differentiate the constraint:

$$\boldsymbol{x} \cdot \dot{\boldsymbol{x}} = 0.$$

Differentiate again:

$$\dot{\boldsymbol{x}}^2 + \boldsymbol{x} \cdot \ddot{\boldsymbol{x}} = 0.$$

Using $m\ddot{\boldsymbol{x}} = 2\lambda\boldsymbol{x}$,

$$\dot{\boldsymbol{x}}^2 + \frac{2\lambda}{m}\boldsymbol{x}^2 = 0.$$

Since $\boldsymbol{x}^2 = R^2$,

$$\lambda = -\frac{m\dot{\boldsymbol{x}}^2}{2R^2}.$$

Thus

$$\boxed{m\ddot{\boldsymbol{x}} = -m\frac{\dot{\boldsymbol{x}}^2}{R^2}\boldsymbol{x}.}$$

The force is inward and supplies the centripetal acceleration required by the constraint.

7 Symmetries and Noether's theorem

7.1 Cyclic coordinates

If L does not depend on a coordinate q^k, then

$$\frac{\partial L}{\partial q^k} = 0.$$

The Euler-Lagrange equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^k}\right) = 0.$$

Therefore the conjugate momentum

$$p_k = \frac{\partial L}{\partial \dot{q}^k}$$

is conserved.

Such a coordinate is called cyclic or ignorable.

7.2 Noether's theorem: derivation

Consider an infinitesimal transformation

$$q^i(t) \mapsto q^i(t) + \varepsilon X^i(q,t), \qquad t \mapsto t + \varepsilon\tau(q,t).$$

For simplicity, first take $\tau = 0$. Suppose the Lagrangian changes by a total derivative:

$$\delta L = \varepsilon \frac{\mathrm{d}F}{\mathrm{d}t}.$$

Under an arbitrary variation,

$$\delta L = \frac{\partial L}{\partial q^i}\delta q^i + \frac{\partial L}{\partial \dot{q}^i}\delta \dot{q}^i.$$

Using integration by parts at the level of the integrand,

$$\delta L = \left(\frac{\partial L}{\partial q^i} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}^i}\right)\delta q^i + \frac{\mathrm{d}}{\mathrm{d}t}\left(p_i \delta q^i\right).$$

On shell, the Euler-Lagrange term vanishes, so

$$\delta L = \frac{\mathrm{d}}{\mathrm{d}t}(p_i \delta q^i).$$

If also $\delta L = \varepsilon \frac{\mathrm{d}F}{\mathrm{d}t}$, then

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(p_i X^i - F\right) = 0.$$

Thus

$$\boxed{Q = p_i X^i - F}$$

is conserved.

Noether's theorem, elementary form

If a continuous transformation changes the Lagrangian at most by a total time derivative, then there is a conserved quantity. Symmetry of the action, not necessarily strict invariance of the Lagrangian, is the essential condition.

7.3 Time translation and energy

If $L(q, \dot{q})$ has no explicit time dependence, then energy is conserved. Define

$$E = \dot{q}^i p_i - L, \qquad p_i = \frac{\partial L}{\partial \dot{q}^i}.$$

Differentiate:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \ddot{q}^i p_i + \dot{q}^i \dot{p}_i - \left(\frac{\partial L}{\partial q^i} \dot{q}^i + \frac{\partial L}{\partial \dot{q}^i} \ddot{q}^i + \frac{\partial L}{\partial t} \right).$$

Since $p_i = \partial L / \partial \dot{q}^i$, the first and third terms cancel. Using $\dot{p}_i = \partial L / \partial q^i$, the remaining coordinate terms cancel. Therefore

$$\frac{\mathrm{d}E}{\mathrm{d}t} = -\frac{\partial L}{\partial t}.$$

If $\partial L / \partial t = 0$, then

$$\boxed{E = \dot{q}^i p_i - L = \text{constant}.}$$

7.4 Spatial translations and momentum

For a particle in three dimensions,

$$L = \frac{1}{2} m \dot{\boldsymbol{x}}^2 - V(\boldsymbol{x}).$$

If V is invariant under translations in the x direction, then $\partial L / \partial x = 0$ and

$$p_x = m\dot{x}$$

is conserved. If V is independent of all components of $\boldsymbol{x}$, then full linear momentum

$$\boldsymbol{p} = m\dot{\boldsymbol{x}}$$

is conserved.

7.5 Rotations and angular momentum

Under an infinitesimal rotation with constant vector $\boldsymbol{\theta}$,

$$\delta \boldsymbol{x} = \boldsymbol{\theta} \times \boldsymbol{x}.$$

For a rotationally invariant Lagrangian, Noether's theorem gives

$$Q = \boldsymbol{p} \cdot (\boldsymbol{\theta} \times \boldsymbol{x}) = \boldsymbol{\theta} \cdot (\boldsymbol{x} \times \boldsymbol{p}).$$

Since $\boldsymbol{\theta}$ is arbitrary, the conserved vector is

$$\boxed{\boldsymbol{J} = \boldsymbol{x} \times \boldsymbol{p}.}$$

This is angular momentum.

8 A complete central-force example

Consider a particle of mass m in a central potential $V(r)$. In plane polar coordinates,

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2),$$

so

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r).$$

The coordinate θ is cyclic, hence

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} = \ell$$

is conserved. This is angular momentum.

The radial equation follows from

$$\frac{\partial L}{\partial r} = mr\dot{\theta}^2 - V'(r), \qquad \frac{\partial L}{\partial \dot{r}} = m\dot{r}.$$

Thus

$$m\ddot{r} = mr\dot{\theta}^2 - V'(r).$$

Using $\dot{\theta} = \ell/(mr^2)$,

$$m\ddot{r} = \frac{\ell^2}{mr^3} - V'(r).$$

This can be interpreted as one-dimensional motion in the effective potential

$$V_{\text{eff}}(r) = V(r) + \frac{\ell^2}{2mr^2}.$$

Indeed,

$$m\ddot{r} = -\frac{\mathrm{d}}{\mathrm{d}r}V_{\text{eff}}(r).$$

Circular orbits

Circular orbits satisfy $\dot{r} = 0$ and $\ddot{r} = 0$. Therefore

$$\frac{\ell^2}{mr^3} = V'(r).$$

For the gravitational potential $V(r) = -GMm/r$,

$$V'(r) = \frac{GMm}{r^2}.$$

So

$$\frac{\ell^2}{mr^3} = \frac{GMm}{r^2},$$

which gives

$$\ell^2 = GMm^2r.$$

Since $\ell = mr^2\dot{\theta} = mrv$ for a circular orbit,

$$m^2r^2v^2 = GMm^2r,$$

so

$$\boxed{v^2 = \frac{GM}{r}.}$$

9 Hamiltonian formulation from the action

9.1 Legendre transform

Define canonical momenta

$$p_i = \frac{\partial L}{\partial \dot{q}^i}.$$

If the velocity-momentum relation can be inverted, one defines the Hamiltonian

$$H(q, p, t) = p_i \dot{q}^i - L(q, \dot{q}, t),$$

where $\dot{q}$ is expressed in terms of (q, p, t).

The differential of H is

$$\mathrm{d}H = \dot{q}^i \,\mathrm{d}p_i + p_i \,\mathrm{d}\dot{q}^i - \frac{\partial L}{\partial q^i}\,\mathrm{d}q^i - \frac{\partial L}{\partial \dot{q}^i}\,\mathrm{d}\dot{q}^i - \frac{\partial L}{\partial t}\,\mathrm{d}t.$$

Since $p_i = \partial L/\partial \dot{q}^i$, the $\mathrm{d}\dot{q}^i$ terms cancel. Therefore

$$\mathrm{d}H = \dot{q}^i \,\mathrm{d}p_i - \frac{\partial L}{\partial q^i}\,\mathrm{d}q^i - \frac{\partial L}{\partial t}\,\mathrm{d}t.$$

Using Euler-Lagrange,

$$\frac{\partial L}{\partial q^i} = \dot{p}_i.$$

Hence

$$\frac{\partial H}{\partial p_i} = \dot{q}^i, \qquad \frac{\partial H}{\partial q^i} = -\dot{p}_i, \qquad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.$$

Thus

$$\boxed{\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}.}$$

9.2 Hamilton's principle in phase space

The phase-space action is

$$\mathcal{S}[q, p] = \int_{t_1}^{t_2} \Big(p_i \dot{q}^i - H(q, p, t)\Big)\,\mathrm{d}t.$$

Varying p_i gives

$$\delta_p \mathcal{S} = \int_{t_1}^{t_2} \left(\dot{q}^i - \frac{\partial H}{\partial p_i}\right) \delta p_i \,\mathrm{d}t.$$

Since δp_i is arbitrary,

$$\dot{q}^i = \frac{\partial H}{\partial p_i}.$$

Varying q^i gives

$$\delta_q \mathcal{S} = \int_{t_1}^{t_2} \left(p_i \delta \dot{q}^i - \frac{\partial H}{\partial q^i} \delta q^i\right)\,\mathrm{d}t.$$

Integrating by parts and using fixed q endpoints,

$$\delta_q \mathcal{S} = \int_{t_1}^{t_2} \left(-\dot{p}_i - \frac{\partial H}{\partial q^i}\right) \delta q^i \,\mathrm{d}t.$$

Therefore

$$\dot{p}_i = -\frac{\partial H}{\partial q^i}.$$

9.3 Poisson brackets

For functions $F(q,p)$ and $G(q,p)$ on phase space, define

$$\{F,G\} = \frac{\partial F}{\partial q^i}\frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q^i}.$$

Hamilton's equations imply

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \{F,H\} + \frac{\partial F}{\partial t}.$$

A quantity F is conserved if

$$\{F,H\} + \frac{\partial F}{\partial t} = 0.$$

This is the bridge to canonical quantization, where Poisson brackets are replaced by commutators.

10 Hamilton-Jacobi theory

Hamilton-Jacobi theory rewrites mechanics in terms of a generating function $S(q,t)$, called Hamilton's principal function. It satisfies

$$\boxed{\frac{\partial S}{\partial t} + H\left(q, \frac{\partial S}{\partial q}, t\right) = 0.}$$

The momentum is

$$p_i = \frac{\partial S}{\partial q^i}.$$

10.1 Free particle

For a free particle in one dimension,

$$H = \frac{p^2}{2m}.$$

Hamilton-Jacobi gives

$$\frac{\partial S}{\partial t} + \frac{1}{2m}\left(\frac{\partial S}{\partial x}\right)^2 = 0.$$

Try a separated solution

$$S(x,t) = Wx - Et.$$

Then

$$-E + \frac{1}{2m}(W')^2 = 0.$$

Thus

$$W' = \pm\sqrt{2mE},$$

and

$$S = px - \frac{p^2}{2m}t.$$

The relation $p = \partial S/\partial x$ is immediate.

10.2 Interpretation

Hamilton-Jacobi theory is important because it connects three worlds:

- classical mechanics through canonical transformations,
- geometrical optics through the eikonal equation,
- quantum mechanics through the semiclassical phase $\psi \sim e^{iS/\hbar}$.

11 Fermat's principle as an action principle

In geometrical optics, light rays make the optical path length stationary:

$$\mathcal{S}[\gamma] = \int_\gamma n(\boldsymbol{x})\,\mathrm{d}s,$$

where $n(\boldsymbol{x})$ is the refractive index and $\mathrm{d}s$ is arclength.

If n is constant, stationary paths are straight lines. If n changes across an interface, the variational principle gives Snell's law.

11.1 Derivation of Snell's law

Let light travel from point $A = (x_A, y_A)$ in medium n_1 to point $B = (x_B, y_B)$ in medium n_2, crossing the horizontal interface $y = 0$ at $(x, 0)$. The optical length is

$$\mathcal{S}(x) = n_1\sqrt{(x - x_A)^2 + y_A^2} + n_2\sqrt{(x_B - x)^2 + y_B^2}.$$

Stationarity requires

$$\frac{\mathrm{d}\mathcal{S}}{\mathrm{d}x} = 0.$$

Compute

$$\frac{\mathrm{d}\mathcal{S}}{\mathrm{d}x} = n_1\frac{x - x_A}{\sqrt{(x - x_A)^2 + y_A^2}} - n_2\frac{x_B - x}{\sqrt{(x_B - x)^2 + y_B^2}}.$$

The two fractions are $\sin\theta_1$ and $\sin\theta_2$, where angles are measured from the normal. Hence

$$\boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2.}$$

12 Relativistic particle

The relativistic free-particle action is proportional to proper time:

$$\mathcal{S} = -mc^2 \int \mathrm{d}\tau.$$

In terms of coordinate time t,

$$\mathrm{d}\tau = \sqrt{1 - \frac{v^2}{c^2}}\,\mathrm{d}t,$$

so

$$L = -mc^2\sqrt{1 - \frac{v^2}{c^2}}.$$

The canonical momentum is

$$\boldsymbol{p} = \frac{\partial L}{\partial \boldsymbol{v}} = \frac{m\boldsymbol{v}}{\sqrt{1 - v^2/c^2}} = \gamma m\boldsymbol{v}.$$

The Hamiltonian is

$$H = \boldsymbol{p}\cdot\boldsymbol{v} - L = \gamma mc^2.$$

Using

$$\boldsymbol{p} = \gamma m\boldsymbol{v},$$

one obtains the energy-momentum relation

$$\boxed{E^2 = p^2c^2 + m^2c^4.}$$

12.1 Reparametrization-invariant form

Let a worldline be $x^\mu(\lambda)$. In flat spacetime with signature $(-,+,+,+)$,

$$\mathcal{S} = -mc \int \sqrt{-\eta_{\mu\nu}\dot{x}^\mu \dot{x}^\nu}\, \mathrm{d}\lambda,$$

where dot means $\mathrm{d}/\mathrm{d}\lambda$. This action is invariant under reparametrizations

$$\lambda \mapsto f(\lambda).$$

This invariance is a gauge symmetry: the parameter along the worldline has no direct physical meaning.

13 Charged particle in an electromagnetic field

The action for a particle of charge q in electromagnetic potentials $(\phi, \boldsymbol{A})$ is

$$\mathcal{S} = \int \left[\frac{1}{2}m\dot{\boldsymbol{x}}^2 + q\dot{\boldsymbol{x}} \cdot \boldsymbol{A}(\boldsymbol{x}, t) - q\phi(\boldsymbol{x}, t)\right] \mathrm{d}t.$$

Thus

$$L = \frac{1}{2}m\dot{\boldsymbol{x}}^2 + q\dot{\boldsymbol{x}} \cdot \boldsymbol{A} - q\phi.$$

Compute the canonical momentum:

$$\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{x}}} = m\dot{\boldsymbol{x}} + q\boldsymbol{A}.$$

This is not the same as mechanical momentum $m\dot{\boldsymbol{x}}$.

For the i-th component,

$$\frac{\partial L}{\partial \dot{x}^i} = m\dot{x}^i + qA_i.$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}^i} = m\ddot{x}^i + q\frac{\partial A_i}{\partial t} + q\dot{x}^j \frac{\partial A_i}{\partial x^j}.$$

Also,

$$\frac{\partial L}{\partial x^i} = q\dot{x}^j \frac{\partial A_j}{\partial x^i} - q\frac{\partial \phi}{\partial x^i}.$$

The Euler-Lagrange equation gives

$$m\ddot{x}^i = q\left[-\frac{\partial \phi}{\partial x^i} - \frac{\partial A_i}{\partial t} + \dot{x}^j\left(\frac{\partial A_j}{\partial x^i} - \frac{\partial A_i}{\partial x^j}\right)\right].$$

Using

$$\boldsymbol{E} = -\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t}, \qquad \boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A},$$

this becomes

$$\boxed{m\ddot{\boldsymbol{x}} = q(\boldsymbol{E} + \dot{\boldsymbol{x}} \times \boldsymbol{B}).}$$

This is the Lorentz force law.

13.1 Gauge invariance

Under a gauge transformation

$$\boldsymbol{A} \mapsto \boldsymbol{A} + \boldsymbol{\nabla}\chi, \qquad \phi \mapsto \phi - \frac{\partial \chi}{\partial t},$$

the interaction part of the Lagrangian changes as

$$q\dot{\boldsymbol{x}} \cdot \boldsymbol{\nabla}\chi + q\frac{\partial \chi}{\partial t} = q\frac{\mathrm{d}\chi}{\mathrm{d}t}.$$

Thus L changes by a total derivative. The equations of motion are unchanged.

This is one of the most important lessons of the action principle: gauge-equivalent Lagrangians can differ by total derivatives and still describe the same physics.

14 Classical field theory

14.1 From mechanics to fields

In mechanics, the dynamical variables are functions of time:

$$q^i(t).$$

In field theory, the dynamical variables are fields over spacetime:

$$\phi^A(x),$$

where A labels field components.

The action is

$$\mathcal{S}[\phi] = \int_\Omega \mathcal{L}(\phi^A, \partial_\mu \phi^A, x)\, \mathrm{d}^d x.$$

Here $\mathcal{L}$ is the Lagrangian density.

14.2 Euler-Lagrange equations for fields

Vary

$$\phi^A \mapsto \phi^A + \varepsilon \eta^A.$$

Then

$$\delta\mathcal{S} = \int_\Omega \left(\frac{\partial \mathcal{L}}{\partial \phi^A}\eta^A + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)}\partial_\mu \eta^A \right) \mathrm{d}^d x.$$

Integrating by parts,

$$\delta\mathcal{S} = \int_{\partial\Omega} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)}\eta^A\, \mathrm{d}\Sigma_\mu + \int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi^A} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right) \right] \eta^A\, \mathrm{d}^d x.$$

For variations vanishing on the boundary, the field Euler-Lagrange equations are

$$\boxed{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right) - \frac{\partial \mathcal{L}}{\partial \phi^A} = 0.}$$

15 Example: real scalar field and Klein-Gordon equation

Use units $c = \hbar = 1$ and metric signature $(-, +, +, +)$. A common convention for a real scalar field is

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2.$$

Then

$$\frac{\partial\mathcal{L}}{\partial\phi} = -m^2\phi,$$

and

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = -\partial^\mu\phi.$$

The field Euler-Lagrange equation is

$$\partial_\mu(-\partial^\mu\phi) - (-m^2\phi) = 0.$$

Thus

$$-\partial_\mu\partial^\mu\phi + m^2\phi = 0.$$

Equivalently,

$$\boxed{(\Box - m^2)\phi = 0}$$

with this sign convention if $\Box = \partial_\mu\partial^\mu = -\partial_t^2 + \nabla^2$. Many texts use the equivalent form depending on metric convention.

15.1 Adding interactions

For a ϕ^4 theory,

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4.$$

Then

$$\frac{\partial\mathcal{L}}{\partial\phi} = -m^2\phi - \frac{\lambda}{3!}\phi^3.$$

Therefore

$$\boxed{\Box\phi - m^2\phi - \frac{\lambda}{3!}\phi^3 = 0}$$

under the same sign convention. The exact signs depend on the chosen convention for the metric and Lagrangian density; consistency is more important than notation.

16 Noether theorem in field theory

If a continuous transformation changes the Lagrangian density by a total divergence,

$$\delta\mathcal{L} = \partial_\mu K^\mu,$$

then there is a conserved current

$$\boxed{j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^A)}\delta\phi^A - K^\mu}$$

satisfying

$$\partial_\mu j^\mu = 0$$

on shell.

16.1 Example: complex scalar field and global $U(1)$ symmetry

Let

$$\mathcal{L} = -\partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi.$$

This is invariant under

$$\phi \mapsto e^{i\alpha}\phi, \qquad \phi^* \mapsto e^{-i\alpha}\phi^*.$$

Infinitesimally,

$$\delta\phi = i\alpha\phi, \qquad \delta\phi^* = -i\alpha\phi^*.$$

The current, ignoring the common factor α, is

$$j^\mu = i\left(\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*\right).$$

Thus

$$\boxed{\partial_\mu j^\mu = 0.}$$

The conserved charge is

$$Q = \int j^0 \, \mathrm{d}^3 x.$$

17 Maxwell theory from an action

The electromagnetic field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The Maxwell action with an external current is

$$\mathcal{S}[A] = \int \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - j^\mu A_\mu\right) \mathrm{d}^4 x.$$

Vary A_μ:

$$\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu.$$

Then

$$\delta\left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu}\right) = -\frac{1}{2} F^{\mu\nu} \delta F_{\mu\nu}.$$

Using antisymmetry of $F^{\mu\nu}$,

$$-\frac{1}{2} F^{\mu\nu} \delta F_{\mu\nu} = -F^{\mu\nu} \partial_\mu \delta A_\nu.$$

Thus

$$\delta\mathcal{S} = \int \left[-F^{\mu\nu} \partial_\mu \delta A_\nu - j^\nu \delta A_\nu\right] \mathrm{d}^4 x.$$

Integrate by parts:

$$\delta\mathcal{S} = \int \left[\partial_\mu F^{\mu\nu} - j^\nu\right] \delta A_\nu \, \mathrm{d}^4 x + \text{boundary term}.$$

With suitable boundary conditions, stationarity gives

$$\boxed{\partial_\mu F^{\mu\nu} = j^\nu.}$$

These are the inhomogeneous Maxwell equations. The homogeneous Maxwell equations follow from the definition $F = \mathrm{d}A$, equivalently

$$\partial_{[\lambda} F_{\mu\nu]} = 0.$$

17.1 Gauge invariance and current conservation

Under

$$A_\mu \mapsto A_\mu + \partial_\mu \chi,$$

$F_{\mu\nu}$ is unchanged. The source term changes by

$$-j^\mu \partial_\mu \chi.$$

After integration by parts,

$$-\int j^\mu \partial_\mu \chi \, \mathrm{d}^4 x = \int (\partial_\mu j^\mu) \chi \, \mathrm{d}^4 x + \text{boundary term}.$$

Gauge invariance for arbitrary χ requires

$$\boxed{\partial_\mu j^\mu = 0.}$$

Thus charge conservation is tied to gauge invariance.

18 Yang-Mills action: non-Abelian generalization

For a non-Abelian gauge field,

$$A_\mu = A_\mu^a T^a,$$

where T^a are generators of a Lie algebra. The field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

In components,

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c.$$

The Yang-Mills action is

$$\mathcal{S}_{\mathrm{YM}} = -\frac{1}{4g^2} \int F_{\mu\nu}^a F^{a\mu\nu} \, \mathrm{d}^4 x.$$

The field equation is

$$\boxed{D_\mu F^{\mu\nu} = 0}$$

in the absence of matter, where D_μ is the gauge-covariant derivative.

The essential difference from Maxwell theory is that the gauge field itself carries gauge charge. The equations are nonlinear even in vacuum.

19 Gravity and the Einstein-Hilbert action

19.1 Geodesic action

A free particle in curved spacetime follows a worldline that extremizes proper time:

$$\mathcal{S} = -mc \int \mathrm{d}s = -mc \int \sqrt{-g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu} \, \mathrm{d}\lambda.$$

The resulting Euler-Lagrange equations are geodesic equations:

$$\boxed{\ddot{x}^\rho + \Gamma^\rho_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = f(\lambda) \dot{x}^\rho.}$$

If λ is an affine parameter, then $f(\lambda) = 0$ and

$$\boxed{\ddot{x}^\rho + \Gamma^\rho_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0.}$$

19.2 Einstein-Hilbert action

The gravitational action is

$$\mathcal{S}_{\mathrm{EH}} = \frac{1}{16\pi G}\int R\sqrt{-g}\,\mathrm{d}^4x + \mathcal{S}_{\mathrm{matter}}.$$

Here:

- $g_{\mu\nu}$ is the metric,
- $g = \det(g_{\mu\nu})$,
- R is the Ricci scalar.

Varying the metric gives, after discarding or properly canceling boundary terms,

$$\delta\mathcal{S}_{\mathrm{EH}} = \frac{1}{16\pi G}\int (R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu})\delta g^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x.$$

The matter variation defines the stress-energy tensor:

$$\delta\mathcal{S}_{\mathrm{matter}} = -\frac{1}{2}\int T_{\mu\nu}\delta g^{\mu\nu}\sqrt{-g}\,\mathrm{d}^4x.$$

Stationarity gives

$$\boxed{R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G T_{\mu\nu}.}$$

These are Einstein's field equations.

Boundary terms in general relativity

The variation of the Einstein-Hilbert action contains second derivatives of the metric variation. A well-posed Dirichlet variational problem requires the Gibbons-Hawking-York boundary term when the spacetime boundary is nonempty. Boundary terms are not cosmetic; they are part of the variational principle.

20 Second variation and stability

A stationary path satisfies $\delta\mathcal{S} = 0$. To study stability, compute the second variation. Suppose

$$\mathcal{S}[q + \varepsilon\eta] = \mathcal{S}[q] + \varepsilon\delta\mathcal{S} + \frac{\varepsilon^2}{2}\delta^2\mathcal{S} + O(\varepsilon^3).$$

At a stationary path,

$$\delta\mathcal{S} = 0.$$

The sign of $\delta^2\mathcal{S}$ determines local behavior:

- positive definite: local minimum,
- negative definite: local maximum,
- indefinite: saddle point.

For field theory, the second variation gives the operator governing small fluctuations. Around a classical solution ϕ_{cl},

$$\phi = \phi_{\mathrm{cl}} + \eta.$$

The quadratic part of the action determines perturbative dynamics, propagators, and stability.

21 Semiclassical bridge to quantum mechanics

In the Feynman path integral, the transition amplitude is formally

$$K(q_f, t_f; q_i, t_i) = \int \mathcal{D}q \, \exp\left(\frac{i}{\hbar}\mathcal{S}[q]\right).$$

When $\hbar$ is small compared with the characteristic action scale, rapidly oscillating phases cancel. The dominant contributions come from paths near stationary points of $\mathcal{S}$:

$$\delta\mathcal{S} = 0.$$

Thus classical mechanics emerges as the stationary-phase approximation to quantum mechanics.

Expanding around a classical path,

$$q(t) = q_{\mathrm{cl}}(t) + \eta(t),$$

we get

$$\mathcal{S}[q] = \mathcal{S}[q_{\mathrm{cl}}] + \frac{1}{2}\delta^2\mathcal{S}[\eta] + \cdots.$$

The first variation vanishes. The quadratic fluctuation determinant gives the leading semiclassical prefactor.

22 Worked example: brachistochrone problem

The brachistochrone asks: among curves connecting two points in a vertical plane, along which curve does a particle slide under gravity in the shortest time, assuming no friction?

Let y be downward vertical displacement and x horizontal displacement. Conservation of energy gives

$$\frac{1}{2}mv^2 = mgy,$$

so

$$v = \sqrt{2gy}.$$

The time element is

$$\mathrm{d}t = \frac{\mathrm{d}s}{v} = \frac{\sqrt{1+y'^2}}{\sqrt{2gy}}\,\mathrm{d}x.$$

Thus the functional is

$$T[y] = \int \sqrt{\frac{1+y'^2}{2gy}}\,\mathrm{d}x.$$

Ignoring the constant $1/\sqrt{2g}$, use

$$L(y, y') = \sqrt{\frac{1+y'^2}{y}}.$$

Because L has no explicit x dependence, the Beltrami identity applies:

$$L - y'\frac{\partial L}{\partial y'} = C.$$

Compute

$$\frac{\partial L}{\partial y'} = \frac{y'}{\sqrt{y}\sqrt{1+y'^2}}.$$

Therefore

$$L - y'\frac{\partial L}{\partial y'} = \frac{\sqrt{1+y'^2}}{\sqrt{y}} - \frac{y'^2}{\sqrt{y}\sqrt{1+y'^2}} = \frac{1}{\sqrt{y}\sqrt{1+y'^2}} = C.$$

Thus

$$y(1+y'^2) = a$$

for some constant $a > 0$.

A standard parametrization solving this equation is

$$x = \frac{a}{2}(\theta - \sin\theta), \qquad y = \frac{a}{2}(1-\cos\theta).$$

This is a cycloid. The curve of fastest descent is not a straight line; it is a cycloid.

23 Worked example: catenary

A hanging chain minimizes gravitational potential energy subject to fixed length. Let $y(x)$ describe the chain. The arclength element is

$$\mathrm{d}s = \sqrt{1+y'^2}\,\mathrm{d}x.$$

The potential energy is proportional to

$$U[y] = \int y\sqrt{1+y'^2}\,\mathrm{d}x.$$

The fixed length constraint is

$$\int \sqrt{1+y'^2}\,\mathrm{d}x = \ell.$$

Introduce a multiplier λ and extremize

$$J[y] = \int (y+\lambda)\sqrt{1+y'^2}\,\mathrm{d}x.$$

The effective Lagrangian is

$$L = (y+\lambda)\sqrt{1+y'^2}.$$

Since L has no explicit x dependence, use Beltrami:

$$L - y'\frac{\partial L}{\partial y'} = C.$$

Compute

$$\frac{\partial L}{\partial y'} = (y+\lambda)\frac{y'}{\sqrt{1+y'^2}}.$$

Then

$$L - y'\frac{\partial L}{\partial y'} = \frac{y+\lambda}{\sqrt{1+y'^2}} = C.$$

This gives

$$1 + y'^2 = \frac{(y+\lambda)^2}{C^2}.$$

Let $u = y + \lambda$. Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \sqrt{\frac{u^2}{C^2} - 1}.$$

Solving,

$$\operatorname{arcosh}\left(\frac{u}{C}\right) = \frac{x - x_0}{C}.$$

Therefore

$$\boxed{y(x) = C\cosh\left(\frac{x - x_0}{C}\right) - \lambda.}$$

The hanging chain is a catenary.

24 Worked example: small oscillations from the action

Let a system have Lagrangian

$$L = \frac{1}{2}M_{ij}(q)\dot{q}^i\dot{q}^j - V(q).$$

Suppose q_0 is an equilibrium:

$$\left.\frac{\partial V}{\partial q^i}\right|_{q_0} = 0.$$

Let

$$q^i = q_0^i + \eta^i.$$

To quadratic order,

$$L^{(2)} = \frac{1}{2}M_{ij}(q_0)\dot{\eta}^i\dot{\eta}^j - \frac{1}{2}K_{ij}\eta^i\eta^j,$$

where

$$K_{ij} = \left.\frac{\partial^2 V}{\partial q^i \partial q^j}\right|_{q_0}.$$

The linearized Euler-Lagrange equations are

$$M_{ij}\ddot{\eta}^j + K_{ij}\eta^j = 0.$$

Try normal modes

$$\eta^i(t) = a^i e^{i\omega t}.$$

Then

$$(K_{ij} - \omega^2 M_{ij})a^j = 0.$$

Nontrivial solutions require

$$\boxed{\det(K - \omega^2 M) = 0.}$$

This is the generalized eigenvalue problem for normal-mode frequencies.

25 Worked example: deriving the wave equation from an action

Let $u(x,t)$ be the displacement of a string. The Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\rho u_t^2 - \frac{1}{2}T u_x^2,$$

where ρ is mass density and T is tension.

Compute

$$\frac{\partial \mathcal{L}}{\partial u} = 0, \qquad \frac{\partial \mathcal{L}}{\partial u_t} = \rho u_t, \qquad \frac{\partial \mathcal{L}}{\partial u_x} = -T u_x.$$

The field Euler-Lagrange equation is

$$\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial u_t}\right) + \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial u_x}\right) - \frac{\partial \mathcal{L}}{\partial u} = 0.$$

Thus

$$\rho u_{tt} - T u_{xx} = 0.$$

Therefore

$$\boxed{u_{tt} = c^2 u_{xx}, \qquad c = \sqrt{T/\rho}.}$$

26 Worked example: action for a nonlinear oscillator

Consider

$$L = \frac{1}{2}m\dot{x}^2 - \left(\frac{1}{2}m\omega^2 x^2 + \frac{\lambda}{4}x^4\right).$$

Then

$$\frac{\partial L}{\partial x} = -m\omega^2 x - \lambda x^3, \qquad \frac{\partial L}{\partial \dot{x}} = m\dot{x}.$$

The equation of motion is

$$\boxed{m\ddot{x} + m\omega^2 x + \lambda x^3 = 0.}$$

Energy is conserved because L has no explicit time dependence:

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\omega^2 x^2 + \frac{\lambda}{4}x^4.$$

For small oscillations, the quartic term is negligible and the frequency is approximately ω. For larger amplitudes, the quartic term changes the period. This is a simple classical model behind perturbation theory and interacting field theory.

27 Beltrami identity and first integrals

Many variational problems can be simplified before solving the full Euler-Lagrange equation. Suppose

$$J[y] = \int_{x_1}^{x_2} L(y, y', x)\,\mathrm{d}x.$$

If L has no explicit dependence on x, then the quantity

$$\boxed{L - y'\frac{\partial L}{\partial y'} = \text{constant}}$$

is conserved along an extremal. This is called the Beltrami identity.

27.1 Derivation

Compute

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(L - y'\frac{\partial L}{\partial y'}\right) = \frac{\partial L}{\partial y}y' + \frac{\partial L}{\partial y'}y'' + \frac{\partial L}{\partial x} - y''\frac{\partial L}{\partial y'} - y'\frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial L}{\partial y'}.$$

The y'' terms cancel, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(L - y'\frac{\partial L}{\partial y'}\right) = \frac{\partial L}{\partial x} + y'\left(\frac{\partial L}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial L}{\partial y'}\right).$$

On an extremal, the bracket vanishes. If $\partial L/\partial x = 0$, then

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(L - y'\frac{\partial L}{\partial y'}\right) = 0.$$

Therefore $L - y'\partial L/\partial y'$ is constant.

27.2 Mechanical interpretation

In ordinary mechanics, if $L(q, \dot{q})$ has no explicit time dependence, the conserved quantity is

$$E = \dot{q}^i \frac{\partial L}{\partial \dot{q}^i} - L.$$

The Beltrami identity is the same idea with x playing the role of the independent variable.

28 Maupertuis-Jacobi principle

Hamilton's principle fixes the time interval and extremizes

$$\mathcal{S} = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t.$$

There is another variational principle for conservative systems at fixed energy. Let

$$L = T - V, \qquad T = \frac{1}{2} g_{ij}(q) \dot{q}^i \dot{q}^j,$$

and assume energy

$$E = T + V$$

is fixed. Then physical trajectories in configuration space extremize the abbreviated action

$$W = \int p_i \, \mathrm{d}q^i.$$

Using

$$T = E - V(q),$$

one finds

$$\mathrm{d}s_J^2 = 2(E - V(q)) g_{ij}(q) \, \mathrm{d}q^i \, \mathrm{d}q^j.$$

Thus trajectories at fixed energy are geodesics of the Jacobi metric

$$\boxed{(g_J)_{ij} = 2(E - V) g_{ij}.}$$

28.1 Meaning

Hamilton's principle is a variational principle in spacetime: it asks for $q(t)$. Jacobi's principle is a variational principle in configuration space: it determines the geometric path first, and the time parametrization is recovered afterward from energy conservation.

Free particle as a geodesic

For a free particle, $V = 0$ and $g_{ij} = m\delta_{ij}$. The Jacobi metric is a constant multiple of the Euclidean metric. Its geodesics are straight lines. This reproduces uniform rectilinear motion after time is restored.

29 Functional derivatives and variational notation

For fields, it is often convenient to write the first variation as

$$\delta \mathcal{S} = \int \frac{\delta \mathcal{S}}{\delta \phi(x)} \delta\phi(x) \, \mathrm{d}^d x + \text{boundary terms}.$$

The functional derivative is defined by the coefficient of $\delta\phi(x)$ after all derivatives have been moved away from $\delta\phi$ by integration by parts.

For

$$\mathcal{S}[\phi] = \int \mathcal{L}(\phi, \partial_\mu\phi)\, \mathrm{d}^d x,$$

we have

$$\boxed{\frac{\delta\mathcal{S}}{\delta\phi} = \frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right).}$$

The field equation is then simply

$$\frac{\delta\mathcal{S}}{\delta\phi} = 0.$$

29.1 Example: functional derivative of a gradient energy

Let

$$\mathcal{S}[\phi] = \int \left[\frac{1}{2}(\nabla\phi)^2 + V(\phi)\right] \mathrm{d}^n x.$$

Then

$$\delta\mathcal{S} = \int \left[\nabla\phi\cdot\nabla\delta\phi + V'(\phi)\delta\phi\right] \mathrm{d}^n x.$$

Integrating the first term by parts,

$$\int \nabla\phi\cdot\nabla\delta\phi\, \mathrm{d}^n x = -\int (\nabla^2\phi)\delta\phi\, \mathrm{d}^n x + \text{boundary term}.$$

Therefore

$$\boxed{\frac{\delta\mathcal{S}}{\delta\phi} = -\nabla^2\phi + V'(\phi).}$$

30 Noether's second theorem and gauge redundancy

Noether's first theorem connects global continuous symmetries with conserved currents. Gauge symmetries are different: they depend on arbitrary functions of spacetime. They express redundancy in the description rather than ordinary physical degeneracy.

In Maxwell theory,

$$A_\mu \mapsto A_\mu + \partial_\mu\chi(x).$$

The function $\chi(x)$ is arbitrary. Such a symmetry implies an identity among the equations of motion. For Maxwell theory, the equations of motion are

$$E^\nu \equiv \partial_\mu F^{\mu\nu} - j^\nu = 0.$$

Taking a divergence,

$$\partial_\nu E^\nu = \partial_\nu\partial_\mu F^{\mu\nu} - \partial_\nu j^\nu.$$

The first term vanishes identically because $F^{\mu\nu}$ is antisymmetric while $\partial_\mu\partial_\nu$ is symmetric. Thus gauge invariance is tied to the consistency condition

$$\partial_\nu j^\nu = 0.$$

Noether's second theorem, informal form

A local gauge symmetry implies differential identities among the Euler-Lagrange equations. Gauge symmetry usually means that not all equations are independent; some variables describe redundant coordinates on the space of physical configurations.

31 Singular Lagrangians and constrained Hamiltonian systems

The Legendre transform from $L(q, \dot{q})$ to $H(q, p)$ requires that the Hessian

$$W_{ij} = \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j}$$

be invertible. If W is singular, the theory has constraints. Gauge theories and reparametrization-invariant systems are typically singular.

31.1 Primary constraint: relativistic particle

Use the reparametrization-invariant action

$$\mathcal{S} = -m \int \sqrt{-\dot{x}^2}\, \mathrm{d}\lambda$$

with $c = 1$. The canonical momentum is

$$p_\mu = \frac{\partial L}{\partial \dot{x}^\mu} = m \frac{\dot{x}_\mu}{\sqrt{-\dot{x}^2}}.$$

Therefore

$$p^2 = -m^2.$$

This is not an equation of motion; it is a constraint on phase space:

$$\boxed{p^2 + m^2 = 0.}$$

The canonical Hamiltonian vanishes because the Lagrangian is homogeneous of degree one in velocities:

$$H_c = p_\mu \dot{x}^\mu - L = 0.$$

Dynamics is generated by a constraint Hamiltonian

$$H_T = \frac{e(\lambda)}{2}(p^2 + m^2),$$

where $e(\lambda)$ is a Lagrange multiplier, often called the einbein in related formulations.

31.2 Conceptual lesson

In constrained systems, gauge transformations are often generated by first-class constraints. The true physical phase space is obtained not by simply solving equations of motion but by quotienting out gauge orbits. This is the gateway to Dirac's theory of constraints, BRST quantization, and modern gauge theory.

32 Boundary terms, symplectic potential, and charges

For a field theory, the variation of the action often has the structure

$$\delta\mathcal{S} = \int_\Omega E_A \delta\phi^A\, \mathrm{d}^d x + \int_{\partial\Omega} \Theta(\phi, \delta\phi).$$

Here:

- $E_A = 0$ are the equations of motion,
- Θ is the symplectic potential current integrated over the boundary.

This decomposition is central in the covariant phase-space formalism. The antisymmetrized variation of Θ gives a symplectic current:

$$\omega(\phi; \delta_1\phi, \delta_2\phi) = \delta_1\Theta(\phi, \delta_2\phi) - \delta_2\Theta(\phi, \delta_1\phi).$$

Integrating ω over a Cauchy surface gives the covariant symplectic form. Conserved charges associated with asymptotic symmetries can be derived from this structure.

32.1 Why this matters

For ordinary mechanics, boundary terms are often hidden by fixed endpoint conditions. In gauge theory and gravity, boundary terms determine physical charges: electric charge, ADM energy, angular momentum, black-hole entropy, and central charges in lower-dimensional gravity.

33 Euclidean action and instanton intuition

In quantum theory, one often performs a Wick rotation

$$t = -i\tau.$$

The oscillatory factor

$$e^{i\mathcal{S}/\hbar}$$

becomes, formally,

$$e^{-\mathcal{S}_E/\hbar},$$

where $\mathcal{S}_E$ is the Euclidean action. Stationary points of $\mathcal{S}_E$ can dominate tunneling amplitudes and nonperturbative effects.

33.1 Double-well intuition

For a particle in a potential with two minima, a classical real-time trajectory cannot pass from one minimum to the other if its energy is too low. In Euclidean time, the action is

$$\mathcal{S}_E = \int \left[\frac{1}{2}m\left(\frac{\mathrm{d}x}{\mathrm{d}\tau}\right)^2 + V(x)\right] \mathrm{d}\tau.$$

The Euclidean equation of motion is

$$m\frac{\mathrm{d}^2x}{\mathrm{d}\tau^2} = \frac{\mathrm{d}V}{\mathrm{d}x}.$$

Notice the sign compared with real time. Euclidean classical solutions connecting different minima are instantons. Their action controls exponentially small tunneling factors.

34 Dimensional analysis of the action

The action has dimensions of energy times time:

$$[\mathcal{S}] = [E][t] = [p][q].$$

In quantum mechanics the natural unit of action is Planck's constant $\hbar$. The ratio

$$\frac{\mathcal{S}}{\hbar}$$

is dimensionless and appears in the quantum phase. Classical behavior is expected when typical action scales are large compared with $\hbar$.

In field theory, the action is usually written as

$$\mathcal{S} = \int \mathcal{L}\, \mathrm{d}^d x.$$

Therefore the engineering dimension of $\mathcal{L}$ depends on spacetime dimension. This simple observation is the starting point for power counting and renormalization.

35 Common mistakes and conceptual traps

Mistake 1: confusing δ and d

The symbol d usually describes an ordinary differential, such as a change along a curve. The symbol δ describes a variation of the entire path or field configuration. In mechanics, $\delta q(t)$ is a new function of time, not a small time step.

Mistake 2: forgetting boundary conditions

The integration by parts step produces boundary terms. They vanish only if the chosen variational problem makes them vanish. In field theory and gravity, boundary terms can carry physical information.

Mistake 3: assuming all stationary points are minima

The word "least" is historically important but mathematically misleading. The correct universal condition is $\delta\mathcal{S} = 0$.

Mistake 4: thinking the Lagrangian is unique

Adding a total derivative to L does not change the equations of motion under fixed endpoint variations. Gauge transformations often work exactly this way.

36 Compact reference sheet

Mechanics

$$\mathcal{S}[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t.$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = 0.$$

$$p_i = \frac{\partial L}{\partial \dot{q}^i}, \qquad H = p_i \dot{q}^i - L.$$

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}.$$

$$E = \dot{q}^i p_i - L, \qquad \frac{\mathrm{d}E}{\mathrm{d}t} = -\frac{\partial L}{\partial t}.$$

Fields

$$\mathcal{S}[\phi] = \int \mathcal{L}(\phi^A, \partial_\mu \phi^A, x)\, \mathrm{d}^d x.$$

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} - \frac{\partial \mathcal{L}}{\partial \phi^A} = 0.$$

$$\delta\mathcal{L} = \partial_\mu K^\mu \quad \Longrightarrow \quad j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \delta\phi^A - K^\mu, \qquad \partial_\mu j^\mu = 0.$$

Electromagnetism

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

$$\mathcal{S}[A] = \int \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - j^\mu A_\mu \right) \mathrm{d}^4 x.$$

$$\partial_\mu F^{\mu\nu} = j^\nu.$$

Gravity

$$\mathcal{S}_{\mathrm{EH}} = \frac{1}{16\pi G} \int R\sqrt{-g}\, \mathrm{d}^4 x + \mathcal{S}_{\mathrm{matter}}.$$

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

37 Exercises

1. Derive the Euler-Lagrange equation for a particle with
$$L = \frac{1}{2} m(\dot{x}^2 + \dot{y}^2) - V(x, y).$$
2. Show that if L has no explicit time dependence, then $E = \dot{q}^i p_i - L$ is conserved.
3. For the Lagrangian
$$L = \frac{1}{2} m(\dot{r}^2 + r^2 \dot{\theta}^2) - \frac{1}{2} k r^2,$$
find the conserved angular momentum and the radial equation.
4. Show that adding $\mathrm{d}F(q,t)/\,\mathrm{d}t$ to L does not change the Euler-Lagrange equations for fixed endpoints.
5. Derive the Lorentz force law from
$$L = \frac{1}{2} m \dot{\boldsymbol{x}}^2 + q \dot{\boldsymbol{x}} \cdot \boldsymbol{A} - q\phi.$$
6. Derive the Klein-Gordon equation from
$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2.$$
7. For a complex scalar field, derive the Noether current associated with the global phase symmetry $\phi \mapsto e^{i\alpha}\phi$.
8. Derive Maxwell's equations from the action
$$\mathcal{S}[A] = \int \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - j^\mu A_\mu \right) \mathrm{d}^4 x.$$

9. Compute the second variation for the harmonic oscillator and determine for which time intervals the classical path is a local minimum.
10. Starting from the Einstein-Hilbert action, identify where boundary terms arise in the variation. Explain why the Gibbons-Hawking-York term is needed for a well-posed Dirichlet problem.

38 Selected solutions

Solution to Exercise 1

For

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - V(x, y),$$

we have

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \qquad \frac{\partial L}{\partial x} = -\frac{\partial V}{\partial x}.$$

Thus

$$m\ddot{x} = -\frac{\partial V}{\partial x}.$$

Similarly,

$$m\ddot{y} = -\frac{\partial V}{\partial y}.$$

In vector form,

$$m\ddot{\boldsymbol{x}} = -\boldsymbol{\nabla} V.$$

Solution to Exercise 4

Let

$$L' = L + \frac{\mathrm{d}F}{\mathrm{d}t}.$$

Then

$$\mathcal{S}'[q] = \int_{t_1}^{t_2} L' \,\mathrm{d}t = \mathcal{S}[q] + F(q(t_2), t_2) - F(q(t_1), t_1).$$

For fixed endpoints, the variation of the boundary contribution is zero. Therefore

$$\delta\mathcal{S}' = \delta\mathcal{S}.$$

Both actions give the same equations of motion.

Solution to Exercise 6

For

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2,$$

we compute

$$\frac{\partial \mathcal{L}}{\partial \phi} = -m^2\phi, \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)} = -\partial^\mu\phi.$$

The field Euler-Lagrange equation gives

$$\partial_\mu(-\partial^\mu\phi) + m^2\phi = 0.$$

Thus

$$-\Box\phi + m^2\phi = 0.$$

Depending on the sign convention for $\Box$, this is written as the Klein-Gordon equation.

Closing perspective

The action principle is not just a compact way to derive equations already known from Newtonian mechanics. It is a unifying language. It explains why symmetries imply conservation laws, why gauge freedom is natural, why field equations follow from local densities, why boundary terms matter, and why classical physics appears as a stationary-phase limit of quantum theory.

For theoretical physicists, the action is often the most fundamental object in a model. Once the action is specified, the equations of motion, conservation laws, canonical structure, perturbation theory, and coupling to other fields can often be derived systematically.